Scientific computing II 2026

Exercise 1
Submit your solution to Moodle no later than Thursday 5.11.2026 23:59
Fortran programs `hello.f90` *and* `numbers.f90` *are given in Moodle.*
For all problems, maximum is 2 points.

Problem 1.
Do the course starting quiz on course Moodle page. Note that it is enough to do the quiz to get the 2 points from this exercise. Whether your answers were right or wrong does not affect the points.

Problem 2.
Compile the Fortran program in file `hello.f90` so that the executable's name is '`moi`'. Run the program. Copy-paste the commands you gave and the program output to your solution text.

Problem 3.
Compile and run program `numbers.f90`. What output do you get and why?

Problem 4.
Write a Fortran program that calculates the velocity of the ball v when it hits the earth from a given height h, using the following equation (air resistance in neglected here)

$$v = \sqrt{2gh},$$

where g is the acceleration due to gravity on the surface of Earth. Height h is obtained as input from the user. Use the program to calculate the velocity for a height of 1, 10, and 100 meters. The square root function in Fortran is called `sqrt`.

Problem 5.
The hyperbolic cosine function is defined by the equation

$$\cosh x = \frac{e^x + e^{-x}}{2}.$$

Fortran also has its own function `cosh(x)`. Write a Fortran program where you compare the values obtained from the expression above and from Fortran's `cosh(x)` function. What is the order of magnitude of the relative difference between these two values in the interval $x \in [0, 1]$.

Problem 6.
Write a Fortran program that prints out values for the function $f(t)$ in interval $t \in [-3, 3]$. Redirect the output to a file and plot it with e.g., Python and `matplotlib`. Attach the plot in you solution.

$$f(t) = \begin{cases} 3t^2 + 5 & t < 0 \\ -3t^2 + 5 & t \geq 0 \end{cases}$$

There once were two codes of renown,
Fortran and C++ wore the crown.
One raced through arrays,
One templated ways,
And both made the runtimes go down.